

Data analysis - Cadaver Measurements Current Ratios

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data and Organize it

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Cadaver/")
data <- read.csv("RatioElectrodeData.csv")

# add cortical (contacts 15 - 9) and hipp (contacts 8 - 1) regions
data$contact <- as.numeric(data$contact)
data$region[data$contact>8] <- "Cortex"
data$region[data$contact<9] <- "Hipp"

# Assign factors
data$select <- factor(data$select)
data$type <- factor(data$type)
data$region <- factor(data$region)
data$location <- factor(data$location)

# Extract only "hippocampal" electrodes
Hipp <- subset(data, region=="Hipp")
```

Calculate and Plot Ratios between Ant and Post electrode, Figure 1i

```
library(wesanderson)
library(dplyr)
TI.hipp <- Hipp %>%
  group_by(type) %>%
  mutate(mean_norm = env_amp / mean(env_amp[location == "Post"])*100,
         median_norm = env_amp / median(env_amp[location == "Post"])*100) %>%
  ungroup

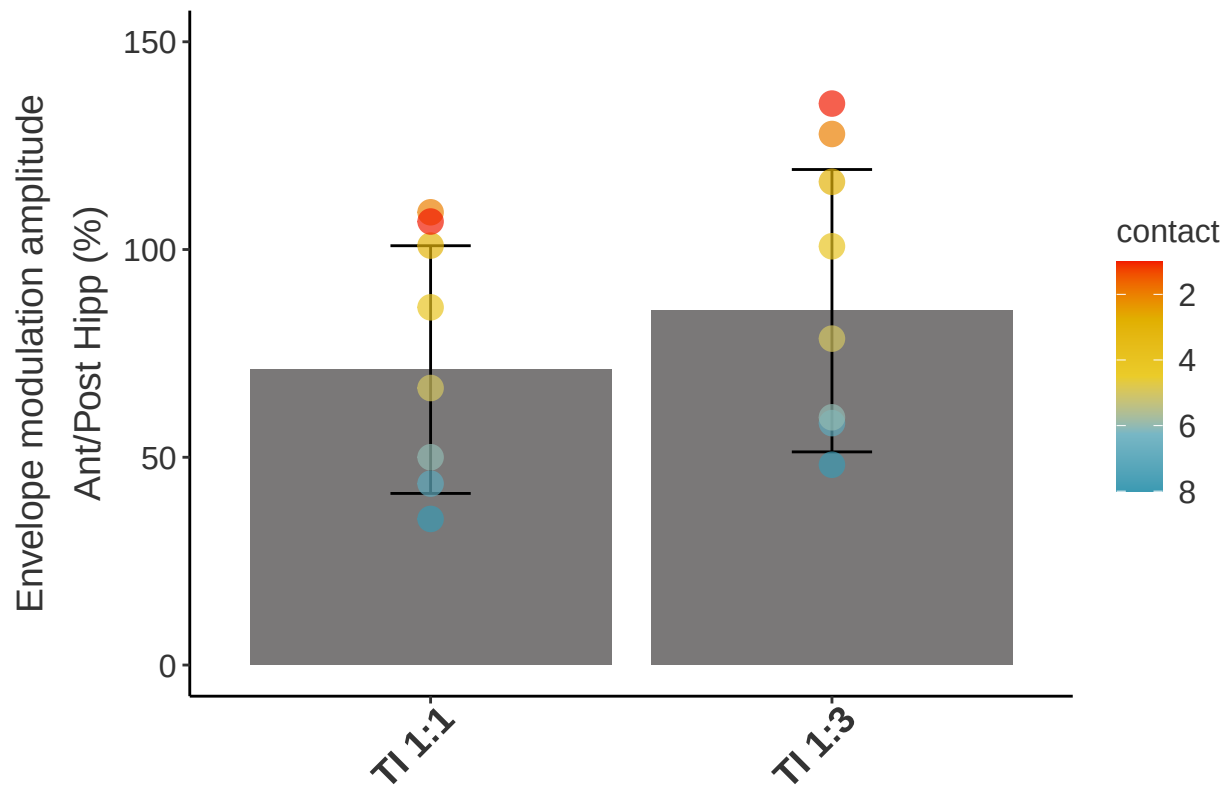
Ant.Post.norm <- subset(TI.hipp, location=="Ant")
```

```

Ant.Post.norm.gr <- ddply(Ant.Post.norm, c("location","type"), summarise,
                          median_env = mean(median_norm),
                          mean_env = mean(mean_norm),
                          sd_env = sd(mean_norm),
                          se_env = se(mean_norm))

Ant.Post.ratio <- ggplot() +
  geom_bar(data = Ant.Post.norm.gr, aes(x=type, y=median_env), stat="identity", fill = "#7a7878") +
  geom_errorbar(data = Ant.Post.norm.gr, aes(x=type, ymin=(median_env-sd_env), ymax=(median_env+sd_env)) +
  geom_point(data = Ant.Post.norm, aes(x=type, y=mean_norm, color=contact), size=4, alpha=0.7) + scale_y_continuous(limits=c(0, 150)) +
  scale_x_discrete(labels=c("TI 1:1", "TI 1:3")) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
        axis.text.x = element_text(color = "grey20", size = 14, angle = 45, hjust=0.9, face = "bold"),
        axis.text.y = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
        axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
        legend.text = element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        legend.background = element_blank(),
        legend.title=element_text(color = "grey20", size = 12, angle = 0, face = "plain"),
        panel.background = element_blank()) +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
        strip.text.y = element_text(size = 14)) + ylab(expression(atop("Envelope modulation amplitude
Ant.Post.ratio

```



```
ggsave(Ant.Post.ratio,file="Figure_11.eps", width = 7, height = 9, dpi = 300, units = "cm", device = "pdf")
```

```
t.test(mean_norm~type, data = Ant.Post.norm, paired=TRUE)
```

```
##
## Paired t-test
##
## data: mean_norm by type
## t = -7.7611, df = 7, p-value = 0.0001106
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -20.60979 -10.98396
## sample estimates:
## mean of the differences
## -15.79688
```

```
t.test(median_norm~type, data = Ant.Post.norm, paired=TRUE)
```

```
##
## Paired t-test
##
## data: median_norm by type
## t = -7.6573, df = 7, p-value = 0.0001204
```

```
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -18.545527  -9.794064
## sample estimates:
## mean of the differences
##                -14.1698
```